Nano-Plasticity of Fused Silica under Coupled Thermo-Mechanical Loading

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Abstract

Understanding the plastic deformation behavior of fused silica at the nanoscale and elevated temperatures is critical for precision manufacturing applications. In this study, we investigate the deformation mechanisms of fused silica under coupled thermo-mechanical loading from room temperature to 1,000 °C. Using nano-ramp scratching and transmission electron microscopy (TEM), we discover that plastic deformation is primarily governed by densification, rather than shear flow or hydrostatic stress variations. The plastically deformed region exhibits a nearly uniform density, with a sharp variation gradient localized at the elastic–plastic boundary. No post-yield work-hardening takes place, indicating the material's elastic/perfectly plastic nature. A temperature-dependent constitutive model is developed to reveal the underlying mechanisms governing the onset and evolution of densification-induced plasticity, providing new insight into its thermally activated behavior and quantitatively capturing the observed experimental trends. Additionally, we find that hardness and yield stress decrease linearly with increasing temperature, while the elastic modulus increases, indicating thermal softening with enhanced stiffness. These findings uncover critical mechanistic insights into the thermally activated deformation of fused silica and related amorphous

materials, establishing an innovative predictive framework for their behavior under nanoscale contact conditions.

Keywords: Fused silica; Constitutive modelling; Densification; Prediction of plasticity; Temperature effect.

1. Introduction

Fused silica, or amorphous silicon dioxide, is a high-purity material renowned for its exceptional thermal, optical and mechanical properties, coupled with remarkable resistance to high temperatures and acid etching [1-4]. Its distinctive features include high ultraviolet transmittance, superior hardness, brittleness and low thermal expansion. These distinct properties make fused silica highly suitable for advanced applications in fields such as aerospace, optical instrumentation, energy harvesting systems, microelectronics and semiconductor manufacturing facilities [5-8]. However, the intrinsic hardness, brittleness, low atomic packing density and high wear resistance of the material make it susceptible to crack formation and subsurface damage during ultraprecision machining processes such as grinding, polishing and etching [9-11]. Nevertheless, the nanoscale deformation mechanisms of fused silica remain poorly understood, especially under coupled thermal-mechanical loading. Therefore, gaining a comprehensive understanding of its plastic deformation behaviour from ambient to elevated temperatures is crucial for optimizing performance and ensuring reliability in advanced applications.

Conventional tensile testing methods, while effective for ductile materials, are unsuitable for brittle materials like fused silica, because these methods can introduce defects during sample preparation and bring about premature failure [12, 13]. Consequently, alternative approaches, including indentation testing, dynamic impact testing and fracture toughness measurements, have been commonly employed for brittle materials, which can provide valuable insights into plastic deformation caused by phase transformations and crack initiation and propagation [14-16]. Hardness testing, which assesses the resistance to plastic deformation through indentation, offers rapid evaluations but presents challenges in interpretation due to the differing strain fields that are observed compared to those under tensile or compressive stresses [17]. To address this issue, Tabor [18, 19] proposed that the hardness of ductile metals correlates with their yield stresses (σ^{YS}) *via* the relation $Hv = C\sigma^{YS}$, where C is typically 3. However, for brittle materials like fused silica, measured hardness values often fall below the predicted due to the combined effects of plastic and elastic strains during indentation [20, 21]. Despite this, the hardness–yield stress relationship for fused silica remains poorly understood. Given its brittleness and the complex influence of densification [22], particularly under varying thermal condition, further investigation is needed to determine whether a similar correlation applies.

Indentation-induced damage in fused silica is significantly influenced by densification [23, 24], a phenomenon particularly relevant in SiO_2-rich materials such as fused silica and BK7 glass, which are widely used in large-aperture optics. The stress contours and maximum principal stress are notably affected by densification [25]. Densification in fused silica refers to the compaction or structural rearrangement of its microstructure, typically triggered by mechanical stress, heat, or pressure. This process becomes particularly evident under indentation or high-pressure loading, where localized densification occurs beneath the applied force, resulting in permanent changes in the density of the material. The degree of densification is influenced by factors such as load, indentation depth and temperature, with higher temperatures facilitating greater atomic mobility and more substantial structural rearrangements.

Characterization techniques such as micro-Raman spectroscopy, Brillouin spectroscopy and Small Angle X-ray Scattering (SAXS) have been employed to map the densification and structural modifications beneath indentation or nano-scratched imprints [26-29]. However, these techniques are limited by relatively low spatial resolution, hindering detailed analysis of densification. To enhance spatial resolution, higher indentation loads are used, though this often results in fracturing or cracking, and complicating interpretation [30, 31]. Therefore, further investigation into the densification behavior of fused silica, especially under varying loading conditions and elevated temperatures, is crucial for improving its mechanical understanding and optimizing its practical applications.

Advanced techniques, such as Transmission Electron Microscopy (TEM) combined with radial distribution function (RDF) analysis, can offer a more refined understanding of deformation under indentation or nano-scratched imprints in amorphous materials. Characterizing the localized atomic structure of these materials is challenging due to the absence of long-range order and the variation of coordination numbers at the atomic scale. RDF, which quantifies the density of surrounding atoms or particles, is a critical structural feature influencing the performance of amorphous materials. RDF can be experimentally obtained through methods such as electron diffraction (ED), neutron diffraction (ND), or X-ray diffraction (XRD), which convert diffraction data into information on nearest-neighbor distances and coordination numbers [29, 32]. However, neutron and XRD techniques provide averaged structural information, which may obscure localized variations in heterogeneous materials. In contrast, the combination of RDF with TEM offers the ability to sample areas on the nanoscale, providing enhanced insight into nanoscale structural variations. Specifically, the peaks observed in RDF contain three critical pieces of information: (a) the position of each peak corresponds to the radius of each coordination shell layer; (b) the integrated area represents the number of atoms within each shell; and (c) the width of the peaks indicates the uncertainty in atomic positions, influenced by both static disorder and thermal dynamic effects. This integrated approach offers a more detailed and spatially resolved characterization of atomic structures, essential for understanding the deformation mechanisms in amorphous materials under external stresses.

To address the challenges in characterizing amorphous materials, this study aims to establish a robust relationship between the hardness and yield stress of fused silica across a broad temperature range from ambient conditions to 1,000 °C. If successful, this relationship will offer deeper insight into the deformation behavior of such materials. A constitutive model will be developed, validated experimentally, to incorporate the effects of densification and to predict the distribution of densification under applied loads, enabling a more accurate understanding of fused silica deformation and similar materials. Additionally, to investigate densification mechanisms at the nanoscale, nano-scratch testing combined with TEM and radial distribution function (RDF) analyses will be conducted to examine localized atomic structural variations and clarify the role of densification in material performance. Collectively, these efforts will contribute to a more comprehensive understanding of the deformation mechanisms in fused silica, a representative amorphous material.

2. Methodology

2.1. Experiment

2.1.1. Deformation Characterization

A comprehensive evaluation of the initial surface characteristics of fused silica is crucial for understanding its intrinsic material behavior. Surface roughness, residual stress and microstructural integrity were systematically analysed using atomic force microscopy (AFM; Dimension Edge PSS, Bruker) for topographical assessment, polarized light microscopy (Fullauto StrainEye, LSM-9001LE, Luceo Co. Ltd, Japan) for residual stress evaluation, and transmission electron microscopy (TEM; Talos F200X, Thermo Fisher) for high-resolution microstructural analysis.

Following ramp-nanoscratching, subsurface deformation mechanisms, including plasticity-induced structural modifications, defect formations and potential phase transformations, were investigated using TEM. Electron-transparent specimens were prepared using focused ion beam (FIB) milling

(FEI Nova Nanolab 200), wherein a protective platinum (Pt) layer was deposited before cross-sectioning to a depth of 5 μm. Final thinning below 100 nm facilitated high-resolution TEM imaging for nanoscale deformation analysis.

2.1.2. Nanoindentation and ramp-nanoscratching

Nanomechanical properties of fused silica were systematically and comprehensively assessed using a Hysitron TI-980 TriboIndenter, integrated with a high-temperature module. This state-of-the-art instrument enabled precise and controlled measurement of the material's mechanical behavior, facilitating nanoindentation experiments over an extensive temperature range from 25 to 800 °C, coupled with a broad spectrum of applied loads ranging from 0.5 to 7 mN. A Berkovich diamond indenter was employed to perform indentation mapping with a 3 × 3 array of data points under each testing condition at the dwell time of 2 s. The high-temperature module ensured accurate and reliable characterization of the material's mechanical properties to thermal variations, thereby providing a detailed and nuanced understanding of its hardness, elastic modulus and other critical nanomechanical properties under varying thermal conditions.

Subsequent to the nanoindentation tests, the hardness (H) was determined using the standard Oliver-Pharr method [33], which correlates the measured indentation depth with the applied load. The hardness was calculated from the peak load ($P_{\max}$) and the indentation depth at maximum load ($h_{\max}$) using the following formula:

$$\text{Hardness (H)} = \frac{P_{\max}}{A}$$

where $P_{\max}$ is the maximum applied load and A is the contact area at maximum indentation depth, which is typically derived based on the geometry of the indenter and the measured indentation depth. Moreover, Young's modulus (E) is calculated as follows:

$$\frac{1}{E_{\mathrm{r}}} = \frac{1-\nu^2}{E} + \frac{1-\nu_{\mathrm{i}}^2}{E_{\mathrm{i}}}$$

where E_r is the reduced modulus, v is the Poisson's ratio of the material being tested, and v_i and E_i are the Poisson's ratio and Young's modulus of the indenter, respectively. By rearranging this equation, Young's modulus of the material (E) can be determined. Further, temperature-dependent Young's modulus of the sample was also confirmed by impulse excitation technique (IMCE RFDT HT1600). A detailed derivation is provided in supplementary Appendix 1. Upon determining the hardness and elastic modulus at varying temperatures, an in-depth analysis was conducted, reported here for the first time, to elucidate the complex interplay between hardness and yield stress in fused silica across an extensive range of operational temperatures. A detailed calculation of yield stress is provided in supplementary Appendix 2. This investigation yields novel and highly significant insights into the temperature-dependent mechanical behavior of fused silica, particularly with respect to the interdependence of its hardness and yield stress. The findings substantially advance the understanding of the material's mechanical performance under varying thermal conditions, providing critical implications for its use in applications where temperature-induced mechanical alterations are of paramount importance.

Subsequently, ramp-nanoscratching experiments were undertaken to investigate the subsurface deformation mechanisms of fused silica, with particular focus on the densification process under varying applied loads and thermal conditions. A series of high-load ramp-nanoscratching tests were conducted using a Berkovich diamond indenter integrated into a multi-function tribometer (MFT-5000, Rtec Instruments), equipped with a high-temperature module. The applied loads in these experiments ranged from 25 to 75 mN, while the temperature conditions were varied between 25 °C and 800 °C. The tests were performed at a controlled sliding speed of 0.2 m/s and a sliding distance of 1 mm. To ensure statistical reliability and reproducibility, five independent scratching tracks were executed for each testing condition. This experimental approach was designed to provide an in-depth understanding of the material removal processes and the associated subsurface deformation mechanisms, particularly the densification behavior of fused silica under different mechanical and thermal loading conditions. The findings from this investigation are anticipated to contribute significantly to advancements in ultra-precision machining and material characterization, offering critical insights into the deformation and densification mechanisms of fused silica at both the micro- and nanoscale.

2.1.3. Atomic-scale Densification

To investigate the densification mechanisms of fused silica at the atomic level, electron diffraction (ED) was employed to obtain the atomic radial distribution function (RDF), a key structural descriptor that provides insights into the material's local atomic ordering and its response to external stresses and thermal treatment. ED is particularly advantageous for characterizing the short- and medium-range order in amorphous materials like fused silica, offering high spatial resolution and sensitivity to subtle structural changes that may occur during densification. The RDF is a fundamental quantity that describes the distribution of interatomic distances in a material, revealing how atomic packing and coordination evolve as the material undergoes deformation. In the context of densification, RDF analysis allows for the quantification of changes in atomic spacing, bond angles and the degree of atomic ordering, particularly within the regions subjected to high mechanical stresses or elevated temperatures. A detailed derivation is provided in supplementary Appendix 3.

During the ED experiments, the fused silica samples were subjected to various loading conditions and temperatures, simulating the applied stresses experienced during the densification process. The resulting electron diffraction patterns were analyzed to extract information about the changes in interatomic distances, coordination numbers and local atomic arrangements. These structural parameters were used to calculate the RDF at different stages of densification, providing a comprehensive view of the material's response to applied loads and thermal treatments.

3. Results and discussion

3.1 Relationship between Hardness and Yield Stress at Elevated Temperature

Figure 1a presents the initial surface morphology and crystallographic structure of fused silica. Atomic force microscopy (AFM) analysis revealed a surface roughness below 3 nm, indicating a nanoscopically smooth topography. Residual stress measurements indicated a uniform distribution across the sample, with an average value of ~22.2 MPa, reflecting the influence of prior processing on the stress state. High-resolution transmission electron microscopy (HRTEM) and associated diffraction patterns confirmed the amorphous nature of the material, with no detectable crystalline phases. Following temperature-dependent nanomechanical testing, scanning electron microscopy (SEM) was used to examine the residual indentation imprints (Figure 1b). The imprint size increased with temperature, coinciding with a reduction in nanohardnes, highlighting the pronounced thermal sensitivity of fused silica's resistance to plastic deformation.

Figure 1c summarizes the temperature-dependent evolution of nanohardness and elastic modulus. Nanohardness exhibited a significant monotonic decrease with increasing temperature, consistent with thermally activated softening. Tests were conducted up to 800 °C due to instrumental limitations, though extrapolation suggests continued degradation at higher temperatures. Specifically, nanohardness decreased from 9.8 ± 0.192 GPa at room temperature to 7.6 ± 0.175 GPa at 400 °C and 5.34 ± 0.192 GPa at 800 °C. Linear regression yielded an excellent fit ($R^2 \approx 0.984$), enabling estimation of hardness at 1,000 °C (~4.41 GPa). In contrast, the elastic modulus increased with temperature, as determined by both nanoindentation and impulse excitation technique (IET). Both methods showed consistent trends, and linear regression ($R^2 \approx 0.985$) confirmed a nearly linear increase in modulus with temperature. This counterintuitive stiffening is attributed to thermally induced atomic rearrangements in the silica network, particularly the lateral migration of oxygen atoms, which relax macroscopic strain and promote quasi-crystalline ordering. This structural stabilization leads to an increase in stiffness, with important implications for high-temperature applications where mechanical integrity is critical.

Direct measurement of yield stress in fused silica is inherently challenging due to its high hardness and brittle nature. However, hardness provides a practical means to estimate yield stress, as both are fundamentally related to plastic deformation. In metals, empirical models such as those proposed by Ashby and Jones [34] and Tabor [19] often approximate yield stress as one-third of hardness. However, this approximation fails for brittle materials, where elastic and plastic deformation are of comparable magnitude—consistent with the high elastic recovery index observed in fused silica. Thus, alternative approaches are necessary.

Fused silica can be well-approximated by an elastic/perfectly plastic model as to be validated by the results and discussions in relation to Figures 2 and 3, wherein plastic flow initiates at the yield point with negligible strain hardening. For such systems, a more appropriate relationship is:

$$\frac{H}{\sigma^{YS}} = \frac{2}{3}\left(1 + \ln\frac{E}{3(1-\nu)\sigma^{YS}}\right)$$

$$= \frac{2}{3}\left(1 - \ln 3(1-\nu)\right) + \frac{2}{3}\ln\frac{E}{\sigma^{YS}}$$

$$= A + B\ln\frac{E}{\sigma^{YS}}$$

where $A = \frac{2}{3}(1 - ln3(1 - \nu))$ is a constant dependent on the Poisson's ratio of materials. From the Poisson's ratio of fused silica ($\nu = 0.148$) obtained through its elastic and shear moduli, the material constants A and B are determined to be 0.041 and 2/3, respectively.

The yield stress of fused silica can be determined by plotting both sides of the equation and identifying the intersection. The resulting yield stress values, together with corresponding hardness values, were used to calculate the correlation factor ($C = H/\sigma^{YS}$). As shown in Figure 1d, yield stress decreased with increasing temperature, while C increased, indicating a greater propensity for plastic deformation under thermal loading. At room temperature, the yield stress was measured at 5.49 ± 0.18 GPa, aligning closely with the finite element analysis results reported by Shim et al. [35] and marginally lower than the constitutive modeling estimations by Gadelrab et al. [36]. The correlation factor (C) at room temperature was determined to be 1.77, significantly lower than the commonly reported value of approximately 3 for metallic materials [19]. Similarly, Feng et al. established a Tabor constant (C) of 1.7 for fused silica at room temperature via finite element analysis [37]. As temperature increased, the yield stress declined linearly, reaching 3.25 ± 0.16 GPa at 500 °C and 1.6 ± 0.1 GPa at 1,000 °C. Correspondingly, the correlation factor increased to 2.19 and 2.75, respectively. The nearly linear increase of C with temperature (Figure 1d) underscores the strong temperature-dependence of the mechanical response of fused silica. These findings highlight the importance of incorporating temperature-sensitive property data into the design of fused silica-based systems for high-temperature environments.

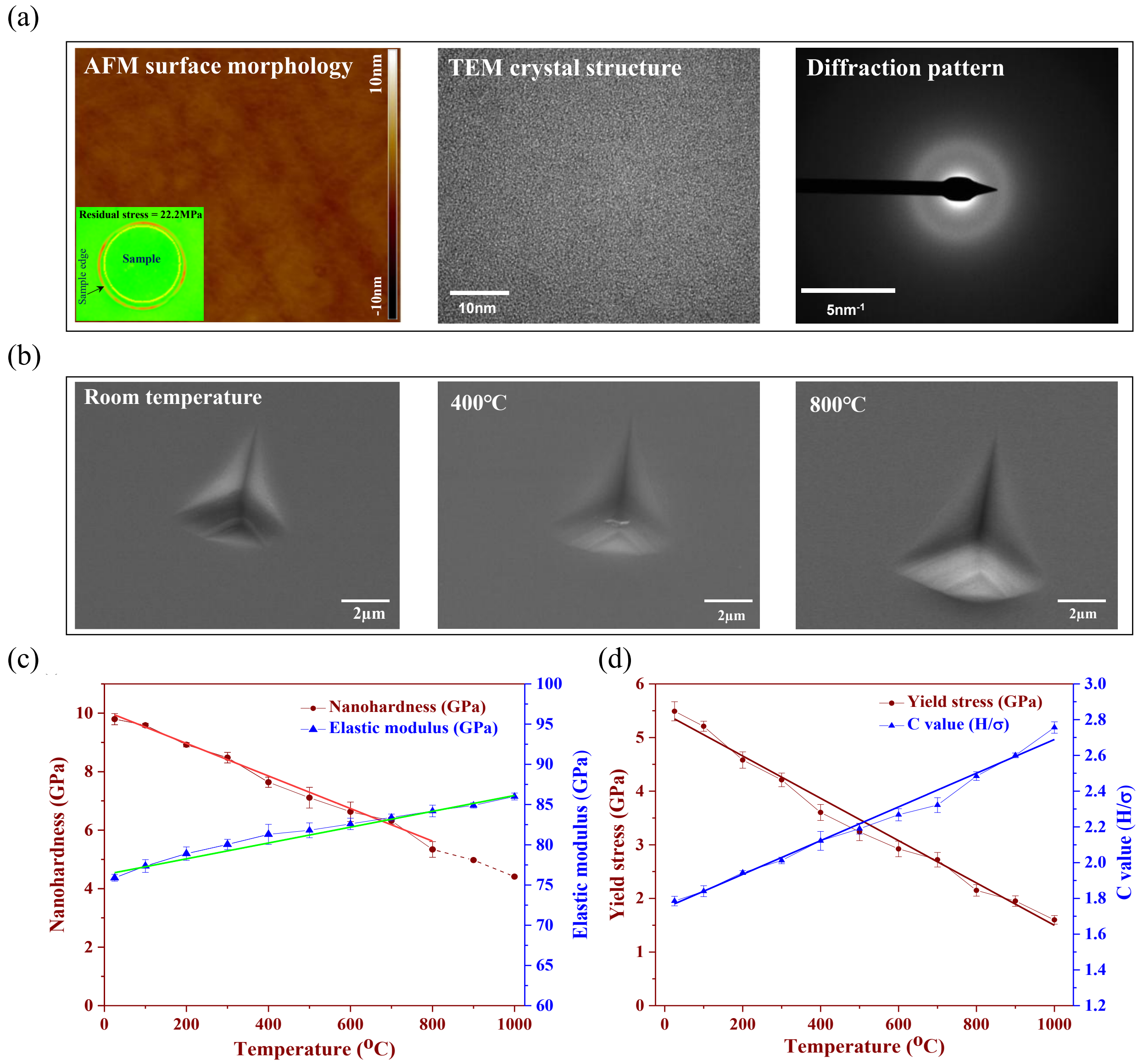

Fig. 1. Physical and nanomechanical characterization of fused silica across varying temperatures. (a) Atomic force microscopy (AFM) image showing nanoscale surface smoothness, complemented by high-resolution transmission electron microscopy (HRTEM) and selected area electron diffraction (SAED) confirming the amorphous atomic structure. (b) Scanning electron microscopy (SEM) images of residual nanoindentation imprints at different temperatures, illustrating thermally induced deformation behavior. (c) Temperature-dependence of nanohardness and elastic modulus, indicating thermal softening and modulus enhancement trends. (d) Yield stress and correlation factor (C), defined as the hardness-to-yield stress ratio) plotted against temperature, revealing the thermal evolution of strength-related parameters. A comprehensive summary of the measured material properties as a function of temperature is provided in Table S1 of the Supplementary Materials.

Figure 2 presents the hardness variation of fused silica under varying indentation loads and temperatures, revealing that the hardness remains nearly constant over the entire load range. This consistent response indicates the absence of work-hardening behavior, which fundamentally differentiates amorphous fused silica from crystalline solids where strain hardening typically arises from the accumulation and interaction of dislocations. In contrast, plastic deformation in fused silica is predominantly governed by irreversible densification and shear flow within the amorphous network. The absence of a load-dependent increase in hardness suggests that once local densification and bond rearrangement have been initiated beneath the indenter, subsequent deformation proceeds through similar structural compaction mechanisms without resistance hardening.

The constant hardness also implies that the structural rearrangements induced during indentation are localized and do not substantially alter the overall resistance to subsequent deformation. This behavior is closely associated with the inability of fused silica to store defects or dislocations, owing to its short-range atomic order and lack of slip systems. Moreover, the deformation is controlled by the breaking and reforming of Si–O–Si bonds, accompanied by volumetric densification in the highly stressed region beneath the indenter tip. Consequently, the mechanical response is dominated by pressure-induced structural collapse rather than strain accumulation, producing a nearly load-insensitive hardness profile.

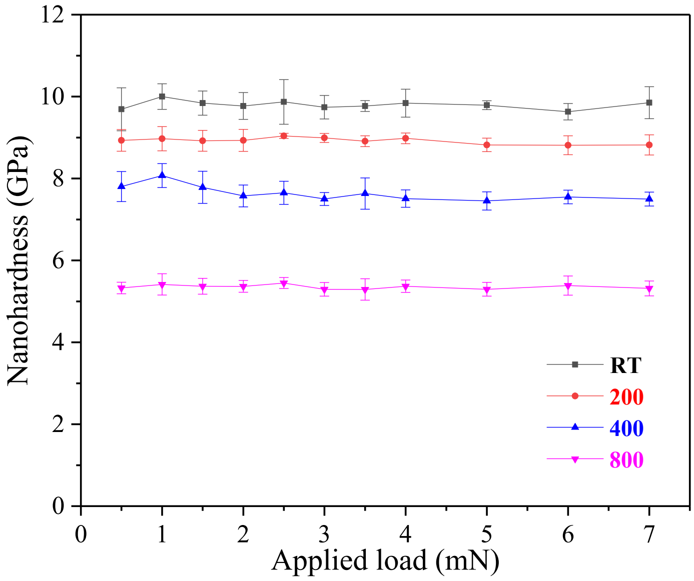

Fig. 2. Load-dependent nanohardness of fused silica measured across varying temperatures. The hardness values were obtained from nanoindentation experiments conducted at applied loads ranging from 0.5 to 7 mN over the temperature range of 25-800 °C, revealing the influence of thermal conditions on the nanomechanical response of fused silica.

3.2 Atomic Scale Quantification of Densification

To elucidate the subsurface deformation mechanisms, a detailed TEM analysis was conducted under varying loads and temperatures after nanoscratching. The corresponding microstructural observations are presented in Figures 3a–c. The micrographs exhibited distinct V-shaped transition boundaries, outlined by white dashed lines, separating the plastically deformed zones (darker contrast) from the surrounding undeformed regions. Electron diffraction patterns confirm that both regions retain an amorphous structure. The darker contrast observed in the deformed zones is indicative of increased densification, attributed to the volumetric compaction of fused silica under applied stresses.

With increasing the normal load, the depth and lateral extent of the deformed region expanded across all tested temperatures. At 400 °C, the extent of deformation slightly increased with load, consistent with thermal softening effects that reduce the hardness of fused silica, underscoring the material's load-dependent mechanical response and highlighting the dominant role of applied stresses in subsurface damage. High-magnification TEM images (Figures 3 a and c) reveal the nucleation of hairline microcracks at the interface between the densified subsurface layer and the underlying matrix. This microcracking suggests a transition in the deformation mode, from densification-driven plastic flow to fracture, once a critical stress threshold is exceeded. Such transitions have important implications for the long-term structural integrity of fused silica, as emerging cracks may evolve into larger-scale failures under continued loading.

The densification behavior of fused silica under coupled thermal–mechanical loading, induced by ramp nano-scratching from room temperature to 800 °C, was characterized using scanning transmission electron microscopy (STEM) combined with nano-beam electron diffraction (NBED). The detailed analytical procedure is provided in Appendix 3.

Figures 3 a–c provide the density distributions in the plastically deformed zones, revealing a consistent pattern even though the temperature at nano-scratching varied. The systematic measurements of densities across a wide range of subsurface from undeformed to deformed areas were conducted at 50 nm intervals from the surface into the substrate, allowing for high-resolution mapping of the structural transformation gradient. These measurements revealed distinct subsurface zones with varying densification levels separated by well-defined boundaries. A key observation across all thermal conditions, *e.g.*, those at room temperature, 400 °C and 800 °C as shown in Figure 3, is the emergence of a distinct densification saturation, highlighted by the dotted lines. It is of significance to note that while the density of the undeformed fused silica, ρ_0, is 2.20 g/cm^3, the density, ρ, across the deformed zone is almost a constant of 2.55 g/cm^3, showing a rather uniform densification ratio of ρ/ρ_0 = 1.16. Yet, it must be highlighted that the variation from ρ_0 to ρ takes place almost immediately across the elastic-plastic boundary. In other words, the densification from ρ_0 to ρ is at a very sharp gradient across the elastic-plastic boundary, beyond which no further compaction occurs even when the mechanical stress and temperature increase. The material within the densified (or plastically deformed) zones remains amorphous, indicating the absence of structural phase changes during coupled thermal-mechanical loading. This distinct behavior, combined with the results of Figure 2 (demonstrating no work hardening), strongly suggests that fused silica is isotropic and homogeneous, and behaves as an elastic/perfectly plastic material under these stressing conditions [38]. Crucially, the densification boundary serves as the initial yield surface, irrespective of temperature. The underlying mechanism for the absence of strain hardening is the uniformity of density across the plastic zone.

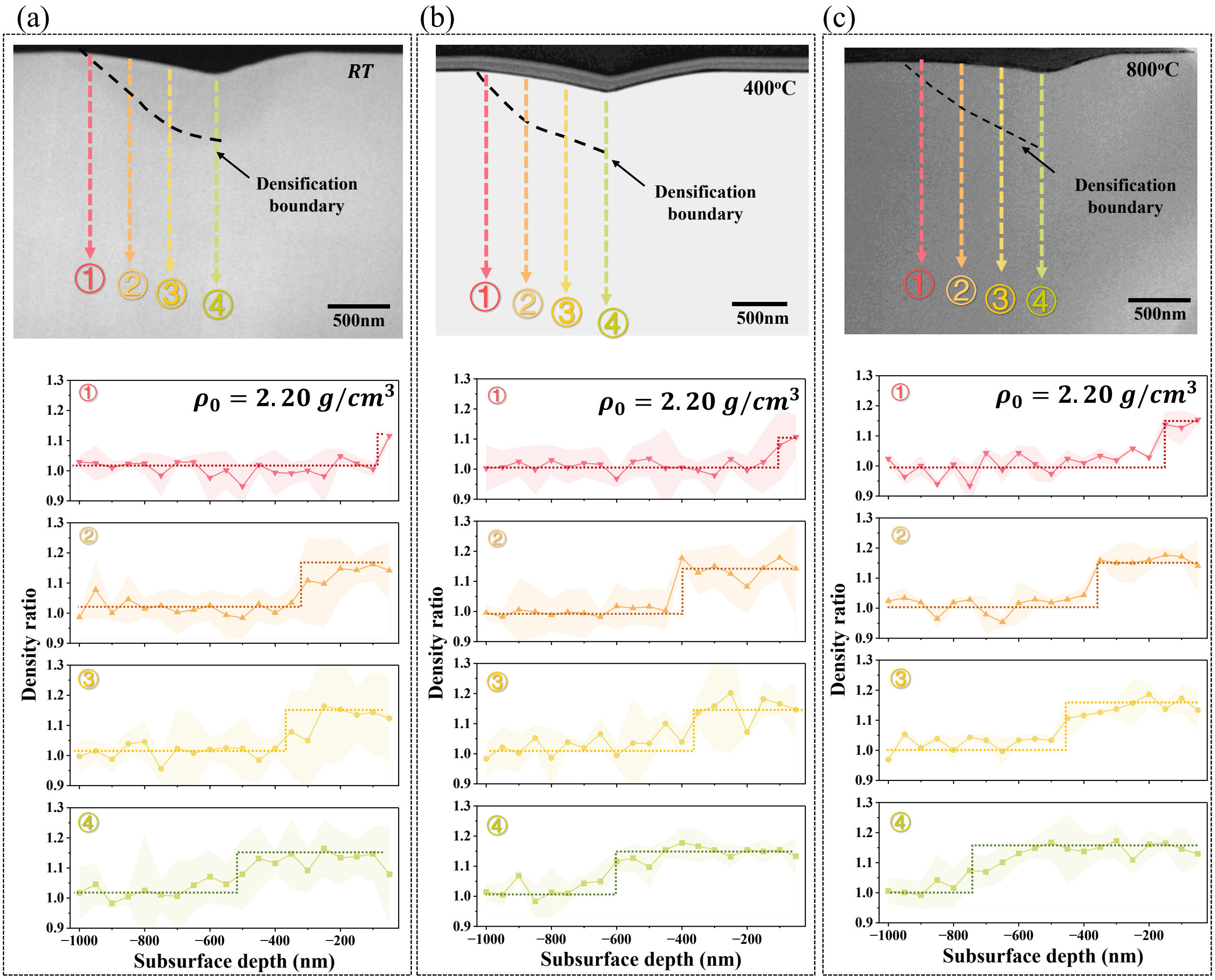

Fig. 3. Nano-beam electron diffraction (NBED) analysis of densification in scratched fused silica under varying thermal conditions and at the normal nano-scratching load of 15 mN. (a)–(c) Cross-sectional bright-field TEM images showing subsurface microstructural variations at (a) room temperature, (b) 400°C, and (c) 800°C, with colored arrows indicating NBED measurement positions. The dashed black lines demarcate the densification threshold boundary, beyond which no further structural compaction occurs. Lower panels present the corresponding density ratio profiles (ρ/ρ_0), where $\rho_0 = 2.20$ g/cm^3 denotes the original density of undeformed fused silica. Shaded regions represent measurement uncertainty. The dotted lines indicate the saturation levels of densification at each position, demonstrating the abrupt transition from densified to undeformed regions and confirming the elastic-perfectly plastic nature of the deformation process. The quantitative densification data measured along the four scan lines (locations 1-4) for each temperature condition are presented in Table S2 in Appendix 3.

The effect of the temperature-dependent behavior of fused silica can also be seen by examining the depths of the densified zones at different temperatures. As shown in Figure 3 a-c, the zone depth was approximately 500 nm at room temperature, spanned to about 600 nm at 400°C, and penetrated to about 750 nm at 800°C. The deeper densification zones at elevated temperatures are due to the enhanced atomic mobility and reduced yield stress at the higher temperatures, facilitating structural reorganization during deformation.

3.3 Residual Plastic Deformation: Theoretical Analysis

3.3.1 Rational of theoretical modelling

The experimental findings above demonstrate that the nanoscopic deformation can be described by the von Mises yielding criterion of plasticity associated with its elastic/perfectly plastic property [38]. The experiment also identifies that the onset of densification corresponds to the onset of initial plastic yielding in the classical theory of plasticity. Here, it must be emphasized that plasticity is irreversible. During plastic deformation, the stress-strain path changes after the material flows, and the stress state upon unloading is not simply the "reverse of the original elastic field." The unloading path depends on the history of accumulated plastic strain and cannot be obtained by a unified linear superposition. In other words, the unloading superposition in the elastic zone is different from that in the plastic zone that has already undergone plasticity. Therefore, the yield boundary after nano-scratching, like those revealed by experiment in Figure 1c, is not merely a function of the stress difference. So, an approach by applying the von Mises criterion after unloading to determine the "residual yield zone" is theoretically incorrect. The true definition of a residual plastic zone should be during the entire loading-unloading history, any region where the equivalent stress has reached the yield point ultimately belongs to the residual plastic zone. This is not necessarily a one-to-one correspondence with the "residual stress field after unloading based on the residual von Mises stress contours." In other words, the plastic zone (material points with plastic strain) will not "disappear" due to unloading (permanent plasticity is retained). After complete unloading, the plastic deformation that has occurred will not disappear; the plastic strain that has been generated is a permanent quantity, and the plastic history of the material point will inevitably be retained. That is to say, the "set of plasticized material points" will not disappear due to elastic rebound (i.e., the plastic zone will not disappear due to unloading). Theoretically, therefore, we can use the von Mises stress criterion to determine the yield surface at nano-scratching loading, but the residual plastically deformed zone (densified zone) after nano-scratching (*i.e.*, after total unloading) must be determined by tracing the residual boundary counter of plastic strain as revealed by the experimental measurement in Figure 3 after nano-scratching.

3.3.2 Implementation

Based on the above strategy, the constitutive behavior of fused silica under elastic – plastic deformation was constructed, as illustrated in Figure 4. All material parameters and indenter geometries were taken from the experiments to ensure consistency. Specifically, the yield stresses were 5.49 GPa at room temperature, 3.60 GPa at 400 °C and 2.26 GPa at 800 °C; and the corresponding Young's moduli were 75.9 GPa, 81.3 GPa and 84.2 GPa, respectively. The indenter geometry and dimensions have been provided in Section 2.1.1. This diagram captures the transition from elastic response to densification-driven plastic flow, illustrating how both temperature-dependent softening and the onset of densification control the subsurface deformation behavior under nanoscale scratching.

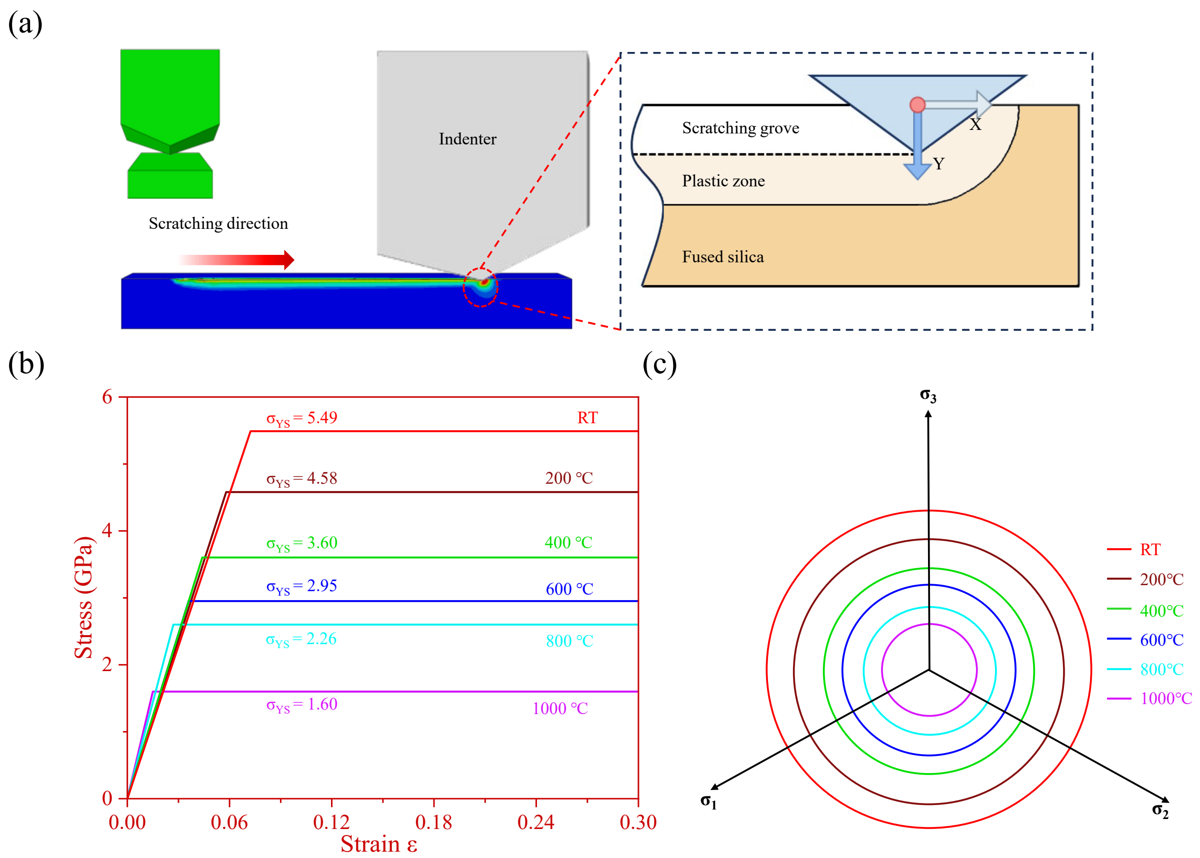

Fig. 4. Constitutive behavior of fused silica: (a) Schematic representation of the stress field distribution, (b) Idealized elastoplastic behavior at varying temperatures, and (c) Evolution of the Π-plane under different temperature conditions.

Figure 4a illustrates the schematic of nano-scratching, where a Berkovich indenter moves along the x-direction. The constitutive behavior of fused silica under coupled thermal–mechanical loading was evaluated using the von Mises criterion, demonstrating an initial linear-elastic response followed by an ideal plastic plateau (Figure 4b). The yield surfaces at different temperatures were determined on the Π-plane in principal stress space (Figure 4c). The results show a systematic reduction in the radius of the yield circles with increasing temperature, reflecting enhanced susceptibility to yielding under elevated thermal conditions. While the yield stress decreases with temperature, promoting earlier onset of plasticity, the elastic modulus simultaneously increases, indicating a complex thermomechanical coupling. This dual effect, thermal softening of the yield strength alongside elastic stiffening, necessitates incorporating temperature-dependent constitutive properties when assessing fused silica's deformation under contact loading. Overall, these findings highlight the critical influence of temperature on the nanoscale mechanical response of fused silica and the importance of capturing thermally driven effects in its constitutive modelling.

3.3.3 Theoretical prediction *versus* experimental measurement

Figures 5a–f shows TEM cross-sections of scratched tracks in fused silica under ramp-nanoscratching at varying loads and temperatures (room temperature and 400 °C). The dotted white curves mark the U-shaped boundaries between deformed (darker contrast) and undeformed regions. Both zones remain amorphous, but the darker contrast in the deformed zones indicates densification, as previously discussed. This contrast arises from local variations in atomic packing density, providing a clear visual marker of stress-affected regions (Figures 5a–f). Quantitative analysis reveals that both scratch depth and the extent of the deformed zone increase with applied load and temperature. At higher loads, the deformed region expands laterally and vertically, with the formation of hairline microcracks (Figures 5c, f), reflecting enhanced structural compaction and localized damage. These observations highlight the load- and temperature-dependent deformation of fused silica, driven by the combined effects of densification and shear flow, and provide direct microstructural evidence of the material's subsurface response under thermo-mechanical loading.

The theoretical analysis based on the above model captures well the evolution of densification-induced plastic deformation in fused silica during both loading and unloading. This predictive capability is supported by experimental evidence which confirms that the as-received specimens contain negligible pre-existing stress prior to nano-scratching, consistent with the assumptions of plasticity theory [38]. Figures 5g−i provide a comparative assessment of the experimentally observed and theoretically predicted elastic−plastic transition boundaries within the subsurface region of fused silica under different scratching loads and temperatures. The experimentally determined boundaries, shown on the right-hand side, are delineated by blue and red curves for room temperature and 400 °C, respectively, while the theoretically predicted boundaries appear on the left-hand side. The close correspondence between experiment and theory demonstrates that the model successfully reproduces the onset and spatial development of plastic deformation driven by densification under coupled thermo-mechanical loading. This agreement highlights that despite the structural complexity of fused silica, its deformation behavior under nanoscratching can be effectively described by classical solid mechanics formulations when the densification effect is properly incorporated.

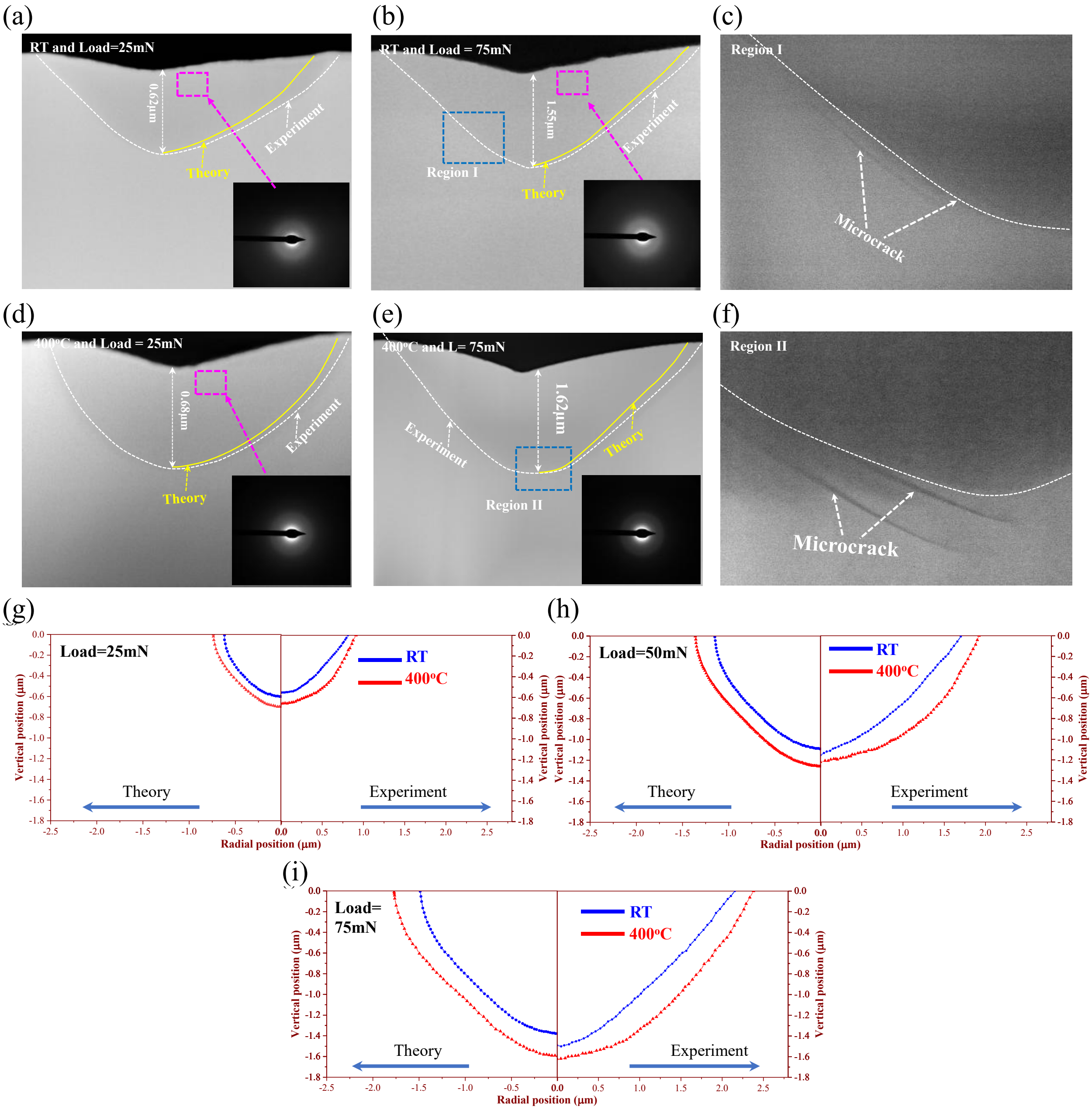

Fig. 5. Subsurface deformation mechanisms of fused silica under ramp-scratching. (a–c) at room temperature and (d–f) at 400 °C. The applied loads are (a, d) 25 mN, and (b, e) 75 mN. Magnified TEM images in (c) and (f) correspond to the marked regions in Figs. 5(b) and 5(e), respectively, providing detailed insights into localized structural transformations. (g–i), Comparison between experimentally observed boundaries separating elastic and densified zones and theoretical predictions based on equivalent plastic strain for 25 mN (g), 50 mN (h), and 75 mN (i) at room temperature and 400 °C.

As well known, hydrostatic stress fundamentally governs volumetric deformation and thus serves as a principal driving force for material densification, as negative hydrostatic stress promotes localized atomic rearrangement and structural compaction. In fused silica, however, although the magnitude of negative hydrostatic stress varies markedly (Figure 6), nanoscopic observations reveal that the density within the densified zone remains nearly constant. This implies that hydrostatic stress contributes to the initiation of densification by facilitating the collapse of open network structures, as one expected, but exerts a negligible influence on its subsequent evolution. Once densification is triggered, the local atomic configuration reaches a mechanically stable state with minimal free volume, rendering the material effectively incompressible. In accordance with solid mechanics theory [38], the Poisson's ratio within the densified region approaches 0.5, confirming this incompressible behavior. These results elucidate the mechanistic transition from stress-assisted structural collapse to incompressible plastic flow in fused silica under coupled thermo-mechanical loading and unloading, providing deeper insight into the role of hydrostatic stress in contact-induced deformation processes relevant to nano-scale surface machining.

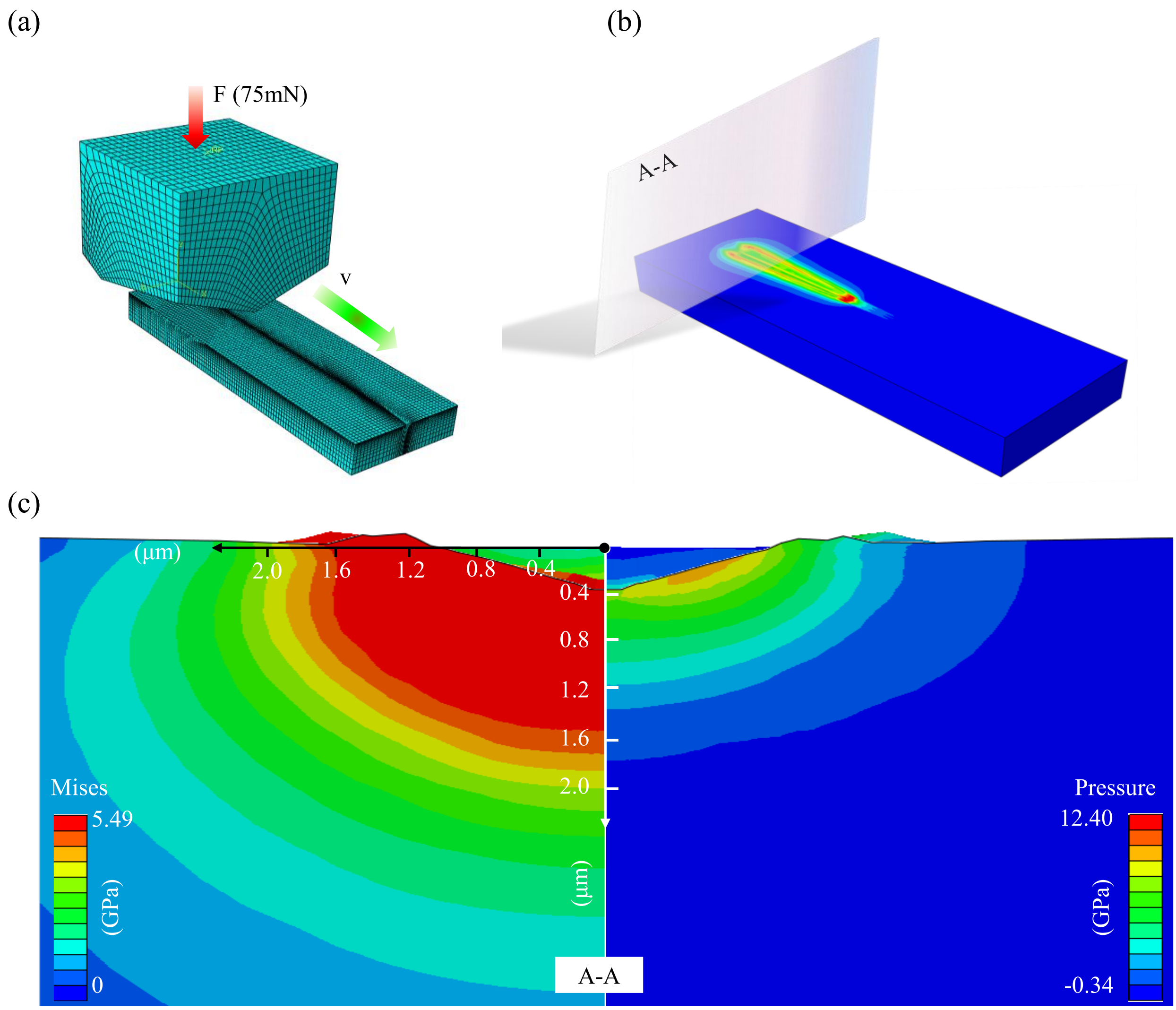

Fig. 6. Stress distribution at 75 mN **revealed by finite element analysis of scratches in fused silica at room temperature**. (a) Finite element analysis (FEA) model for scratch simulation, (b) Deformation diagram of fused silica after scratching, (c) Comparison of von Mises stress (left) and hydrostatic pressure (right) in the scratched region.

4. Conclusions

This study systematically investigated the temperature-dependent nanomechanical response, deformation mechanisms and stress-driven densification behavior of fused silica under complex contact loading and unloading. By integrating nanoindentation, ramp-nanoscratching, advanced electron microscopy and theoretical solid mechanics analysis, several key findings were obtained:

(i) Constitutive behavior and densification mechanism.

Theoretically, fused silica can be modeled as an isotropic, homogeneous, elastic–perfectly plastic material, with plastic deformation governed by densification. Once densification occurs, the material density remains constant and is unaffected by temperature variations between room temperature and 1,000 °C. The onset of densification can be accurately predicted using the von Mises yield criterion. Within the densified zone, fused silica behaves as an incompressible solid with a Poisson's ratio approaching 0.5.

(ii) Temperature-dependent nanomechanical properties.

A pronounced temperature dependence was observed in the nanomechanical properties of fused silica. The nanohardness decreased from 9.8 GPa at room temperature to 4.41 GPa at 1,000 °C, while the elastic modulus increased systematically with temperature, attributed to thermally induced atomic rearrangements within the amorphous network. The yield stress decreased linearly from 5.49 GPa to 1.6 GPa over the same temperature range. Meanwhile, the correlation factor between nanohardness and yield stress, C, increased with temperature, indicating an enhanced tendency toward plastic deformation.

(iii) Thermal influence on yielding behavior.

The yield surface of fused silica in principal stress space progressively contracts with increasing temperature, confirming the dominance of shear-driven plasticity under thermal softening. This finding underscores the critical role of temperature-dependent material properties in governing the nanoscale deformation and densification behavior of fused silica under contact loading.

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